STA 2023 - Final Exam Targeted Study Guide

Spring 2026

Study Topic 1: Foundations of Statistics

Basic Statistical Terminology

Statistics is the science of collecting, describing, and interpreting data. A **population** is the entire group of interest, and a **parameter** describes it. A **sample** is a subset of the population, and a **statistic** describes the sample.

Types of Data

Qualitative (categorical) variables classify individuals based on attributes. **Quantitative** variables provide numerical measures and can be **discrete** (countable) or **continuous** (measurable).

Levels of Measurement

i. **Nominal:** Names or labels.

ii. **Ordinal:** Ordered categories.

iii. **Interval:** Meaningful differences, no true zero.

iv. **Ratio:** Meaningful ratios, true zero exists.

Study Design

Observational studies measure characteristics without influence, showing association. **Designed experiments** apply treatments to observe effects, allowing for causation claims.

Experimental Design

Observational studies cannot prove causation due to **lurking or confounding variables** that may affect the response variable.

Sampling Methods

In a **simple random sample**, every individual and every possible sample of size n has an equal chance of being selected. Use `randInt(min, max, n)` on a calculator to generate.

Study Topic 2: Descriptive Statistics and Visualization

Descriptive Statistics

The **five-number summary** describes a dataset using: Minimum, Q_1, Median, Q_3, and Maximum. Data must be ordered from smallest to largest.

Data Visualization (Boxplot)

A boxplot visually displays the five-number summary. The **Interquartile Range (IQR)** is used to identify outliers.

IQR and Fences

$IQR = Q_3 - Q_1$

Lower Fence $= Q_1 - 1.5(IQR)$

Upper Fence $= Q_3 + 1.5(IQR)$

Shape of Distribution

If the median is closer to Q_1, the distribution is **right-skewed**. If closer to Q_3, it is **left-skewed**.

Study Topic 3: Probability Theory

Basic Probability

Marginal probability is the likelihood of a single event occurring.

Classical Probability

$$P(A) = \frac{n(A)}{N}$$

Probability Rules

The **General Addition Rule** calculates the probability of event A or event B occurring.

General Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Conditional Probability

The probability of event B given that event A has occurred.

Conditional Probability

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Independence of Events

Two events are independent if $P(B|A) = P(B)$ or:

Multiplication Rule for Independent Events

$$P(A \cap B) = P(A) \times P(B)$$

Multiplication Rule

Used for a sequence of dependent events (e.g., without replacement).

General Multiplication Rule

$$P(A \cap B \cap C) = P(A) \times P(B|A) \times P(C|A \cap B)$$

Study Topic 4: Random Variables and Distributions

Discrete Random Variables

A probability distribution lists all outcomes x and their probabilities $P(x)$. Requirements: $0 \leq P(x) \leq 1$ and $\sum P(x) = 1$.

Expected Value

The mean (μ) of a discrete random variable, representing the long-term average.

Expected Value

$$E(X) = \sum[x \times P(x)]$$

Variance and Standard Deviation

Measures the average spread of outcomes around the expected value.

Discrete Standard Deviation

$$\sigma = \sqrt{\sum[x^2 \times P(x)] - \mu^2}$$

Binomial Distribution

Fixed trials n, independent trials, two outcomes, constant probability p.

Binomial Probability

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Binomial Properties

Binomial Mean and SD

$$\mu = n \times p$$

$$\sigma = \sqrt{n \times p \times (1 - p)}$$

Study Topic 5: Sampling Distributions

Sampling Distribution of the Mean

Describes the distribution of all possible sample means.

Mean and Standard Error

$\mu_{\bar{x}} = \mu$

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

Central Limit Theorem

The distribution of $\bar{x}$ is approximately normal if $n \geq 30$ or the population is normal.

Z-score for Means

$$Z = \frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}}$$

Sampling Distribution of Proportion (Conditions)

Normal approximation for $\hat{p}$ requires:

Success-Failure Condition

$n \times p \geq 10$ and $n \times (1 - p) \geq 10$

Sampling Distribution of Proportion (Parameters)

Mean and Standard Error for Proportions

$\mu_{\hat{p}} = p$

$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$

Sampling Distribution of Proportion (Probability)

Calculating the probability of a sample proportion.

Z-score for Proportions

$$Z = \frac{\hat{p} - \mu_{\hat{p}}}{\sigma_{\hat{p}}}$$

Study Topic 6: Inferential Statistics - Confidence Intervals

Confidence Intervals for Proportions

Provides a range of plausible values for the population proportion.

Confidence Interval for Proportion

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Confidence Intervals (Interpretation)

We are $C\%$ confident that the true population parameter falls between the lower and upper bounds of the interval.